

Your grade: 100%

Your latest: 100% > Your highest: 100% > To pass you need at least 80%. We keep your latest score.

Next item →

1. What defines the size of a matrix? 1 / 1 point

- ☒ Number of rows and columns
- ☐ Number of rows only
- ☐ Number of columns only
- ☐ The sum of rows and columns
- ☐ None work

Correct. A matrix is defined by its dimensions, specified by the number of rows and columns.

2. Which of the following is a correct notation for a matrix with 3 rows and 4 columns? 1 / 1 point

- ☐ 4 × 3
- ☐ 3 × 3
- ☐ 3 × 4
- ☐ 4 × 4
- ☒ None work

Correct. This notation represents a matrix with 3 rows and 4 columns.

3. Which of the following matrices can be added together? 1 / 1 point

- ☐ A 3x1 matrix and a 3x1 matrix
- ☐ A 3x3 matrix and a 3x2 matrix
- ☐ A 1x3 matrix and a 3x1 matrix
- ☐ A 3x2 matrix and a 2x3 matrix
- ☒ None work

Correct. The matrices have the same dimensions and can be added.

4. What is the result of multiplying a 2x3 matrix by a scalar 3? 1 / 1 point

- ☐ A 2x3 matrix where each element is tripled
- ☐ A 2x3 matrix where each element is divided by 3
- ☐ A 3x2 matrix where each element is divided by 3
- ☐ A 3x2 matrix where each element is tripled
- ☒ None work

Correct. Multiplying a matrix by a scalar means multiplying each element by that scalar.

5. What does it mean if two matrices A and B can be multiplied together? 1 / 1 point

- ☐ A and B have the same number of columns
- ☐ A has the same number of columns as B has rows
- ☐ A has the same number of rows as B has columns
- ☐ A and B have the same number of rows
- ☒ None work

Correct. For two matrices to be multiplied, the number of columns in the first matrix must equal the number of rows in the second matrix.

6. What is the result of multiplying a 2x3 matrix by a 3x2 matrix? 1 / 1 point

- ☐ A 2x3 matrix
- ☐ A 3x3 matrix
- ☐ A 2x2 matrix
- ☐ A 3x2 matrix
- ☒ None work

Correct. The resulting matrix will have the same number of rows as the first matrix and the same number of columns as the second matrix.

7. Which operation is not commutative? 1 / 1 point

- ☒ Matrix multiplication
- ☐ Scalar multiplication
- ☐ Matrix transposition
- ☐ Matrix addition
- ☐ None work

Correct. Matrix multiplication is not commutative.

8. What is the result of adding two matrices of different dimensions? 1 / 1 point

- ☒ The operation is not possible
- ☐ A matrix with dimensions equal to the difference of the original matrices' dimensions
- ☐ A zero matrix
- ☐ A matrix with dimensions equal to the sum of the original matrices' dimensions
- ☐ None work

Correct. Matrices must have the same dimensions to be added together.

9. What is the dimension of the resulting matrix when a 4x2 matrix is multiplied by a 2x3 matrix? 1 / 1 point

- ☐ 4x6
- ☐ 3x3
- ☐ 2x2
- ☐ 4x3
- ☒ None work

Correct. The resulting matrix has the same number of rows as the first matrix and the same number of columns as the second matrix.

10. Given two matrices: 1 / 1 point

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

$$B = \begin{pmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{pmatrix}$$

Calculate the product AB.

Feedback

The product AB is $\begin{pmatrix} 58 & 64 \\ 139 & 154 \end{pmatrix}$

1. Multiply the first row of A by the first column of B:

$$1 \times 7 + 2 \times 9 + 3 \times 11 = 7 + 18 + 33 = 58$$

2. Multiply the first row of A by the second column of B:

$$1 \times 8 + 2 \times 10 + 3 \times 12 = 8 + 20 + 36 = 64$$

3. Multiply the second row of A by the first column of B:

$$4 \times 7 + 5 \times 9 + 6 \times 11 = 28 + 45 + 66 = 139$$

4. Multiply the second row of A by the second column of B:

$$4 \times 8 + 5 \times 10 + 6 \times 12 = 32 + 50 + 72 = 154$$

The resulting matrix is:

$$\begin{pmatrix} 58 & 64 \\ 139 & 154 \end{pmatrix}$$

11. If $C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $D = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$, calculate CD. 1 / 1 point

Feedback

The product CD is $\begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$

1. Multiply the first row of C by the first column of D:

$$1 \times 2 + 0 \times 4 = 2$$

2. Multiply the first row of C by the second column of D:

$$1 \times 3 + 0 \times 5 = 3$$

3. Multiply the second row of C by the first column of D:

$$0 \times 2 + 1 \times 4 = 4$$

4. Multiply the second row of C by the second column of D:

$$0 \times 3 + 1 \times 5 = 5$$

The resulting matrix is:

$$\begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$$

12. Given the matrices: 1 / 1 point

$$E = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix}$$

$$F = \begin{pmatrix} 1 & 3 \\ 5 & 7 \end{pmatrix}$$

Calculate E + F.

Feedback

The sum $E + F$ is:

$$\begin{pmatrix} 3 & 7 \\ 11 & 15 \end{pmatrix}$$

1. Add the corresponding elements:

$$2 + 1 = 3$$

$$4 + 3 = 7$$

$$6 + 5 = 11$$

$$8 + 7 = 15$$

The resulting matrix is:

$$\begin{pmatrix} 3 & 7 \\ 11 & 15 \end{pmatrix}$$

13. Given the matrices: 1 / 1 point

$$G = \begin{pmatrix} 1 & 3 \\ 5 & 4 \end{pmatrix}$$

$$H = \begin{pmatrix} 2 & 6 \\ 3 & 8 \end{pmatrix}$$

Calculate G + H.

Feedback

The sum $G + H$ is:

$$\begin{pmatrix} 3 & 9 \\ 8 & 12 \end{pmatrix}$$

1. Add the corresponding elements:

$$1 + 2 = 3$$

$$3 + 6 = 9$$

$$5 + 3 = 8$$

$$4 + 8 = 12$$

The resulting matrix is:

$$\begin{pmatrix} 3 & 9 \\ 8 & 12 \end{pmatrix}$$

14. Given the matrices: 1 / 1 point

$$I = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 1 & 0 \end{pmatrix}$$

$$J = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

Calculate the product IJ.

Feedback

The product IJ is:

$$\begin{pmatrix} 5 & 14 \\ 5 & 17 \end{pmatrix}$$

1. Multiply the first row of I by the first column of J:

$$2 \times 1 + 0 \times 2 + 1 \times 3 = 2 + 0 + 3 = 5$$

2. Multiply the first row of I by the second column of J:

$$2 \times 4 + 0 \times 5 + 1 \times 6 = 8 + 0 + 6 = 14$$

3. Multiply the second row of I by the first column of J:

$$3 \times 1 + 1 \times 2 + 0 \times 3 = 3 + 2 + 0 = 5$$

4. Multiply the second row of I by the second column of J:

$$3 \times 4 + 1 \times 5 + 0 \times 6 = 12 + 5 + 0 = 17$$

The resulting matrix is:

$$\begin{pmatrix} 5 & 14 \\ 5 & 17 \end{pmatrix}$$